



ICP Final Project Report

Project Title: Hogwarts Sorting System

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1. Abstract:

The Hogwarts Sorting System is a simple C programming project designed to sort students into different Hogwarts houses based on their answers to personality questions. The program asks each student for their name and gives them two questions with four choices each. Each choice has a different score, and the total score determines whether the student belongs to Gryffindor, Ravenclaw, Hufflepuff, or Slytherin. The program also allows the user to display all students, search for a particular student, and find the student with the highest magic score. The project demonstrates the use of fundamental C programming concepts, including arrays, functions, loops, conditional statements, strings, and string comparison and copying functions.

2. Introduction:

The idea of this program is based on the Harry Potter Hogwarts house-sorting concept. In Hogwarts, students are sorted into different houses based on their personalities and qualities. This project uses a similar idea but applies it through a simple scoring system in C. Instead of using complex methods, the program gives different points to each answer and uses the final score to decide the student's house. This makes the project both interesting and useful for demonstrating basic C programming concepts.

3. Methodology /Implementation:

The program starts by creating three global arrays: name stores the names of the students, magic stores their scores, and house stores their assigned Hogwarts houses. The SIZE constant is set to 5, so the program can store information for five students. The addStudents() function is used to collect the student's name and answers. Each answer gives a certain number of points, and the points are added to the student's total score. After calculating the total score, if-else statements are used to determine the student's house. If the score is 80 or above, the student is assigned to Gryffindor. Scores from 60 to 79 result in Ravenclaw, scores from 40 to 59 result in Hufflepuff, and scores below 40 result in Slytherin. The strcpy() function from the string.h library is used to store the house name in the house array. The display() function is used to show all the students along with their scores and houses. A for loop goes through the arrays and prints the stored information. The search() function allows the user to enter a student's name and find their information. It uses the strcmp() function to compare the entered name with the names stored

in the array. If a matching name is found, the student's score and house are displayed. The `topWizard()` function finds the student with the highest magic score. It starts by assuming the first student has the highest score and then compares the remaining students using a for loop. If a higher score is found, the position of that student is stored. Finally, the program displays the name and score of the student with the highest score. The `main()` function controls the entire program through a menu-driven system. A while loop keeps the program running until the user selects option 5. The user can choose to add students, display students, search for a student, or find the top wizard. If-else statements are used to perform the selected operation, making the program easy to navigate and understand.

4. Results / Output:

5. Challenges and Limitations:

As there were several functions and loops to work with, one of the challenges we faced was maintaining the correct placement of brackets and setting the correct range and number of iterations for the loops. Since the program contains multiple functions, it was sometimes difficult to keep track of where each function started and ended. Another challenge was using string functions such as `strcmp()` and `strcpy()`, as this was one of the first times we had used these functions in a big code. The most important limitation we faced was using the `scanf("%s")` function to input student names. We could not use `fgets()` because it made the code more

complicated for us at this stage. As a result, the program can only store student names without spaces, meaning that a full name containing multiple words cannot be entered properly.

6. Conclusion and Discussion

The Hogwarts Sorting System successfully demonstrates how basic C programming concepts can be combined to create an interactive application. The project uses arrays, functions, loops, conditional statements, strings, and a menu-driven structure. The program successfully calculates student scores, assigns Hogwarts houses, displays student information, searches for students, and identifies the Top Wizard. Overall, the project provides a creative way to demonstrate fundamental programming concepts while creating an engaging and interactive system.

7. Future Work:

There are several ways the program could be improved in the future. The number of students could be increased or made dynamic so that the user is not limited to five students. More questions could also be added to make the house selection more accurate and interesting. The program could be improved to allow full names with spaces by using `fgets()` instead of `scanf("%s")`. Additional features such as editing or deleting student records, sorting students by their scores, displaying the number of students in each house, and saving the information to a file could also be added. These improvements would make the system more complete and closer to a real student management program.

8. References:

No external sources were used for the implementation of this program. The project was developed using the fundamental concept of the C programming language.

9. Individual Contributions:

Zaheen

Developed the main structure of the Hogwarts Sorting System. Worked on the student input and score calculation. Implemented the Hogwarts house assignment system.

Liyana

Worked on the display and search functionalities. Tested the program with different student

inputs. Helped identify and fix errors in the program.

Fabliha

Worked on the Top Wizard functionality and menu system. Assisted with program testing and output verification. Helped prepare the final project report and documentation.